Zeina BOURHANE

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Scientist in microbial ecology, I investigate microbial community dynamics, functions and products in extreme environments. My work supports innovative strategies for the sustainable management and conservation of natural resources.

I. Applicant information

Date, place of Birth: 1 January 1990, Darine **Citizenship:** French

LinkedIn: www.linkedin.com/in/zeina-bourhane-772373144 **ORCID:** 0000-0002-9663-2584

II. Academic position

[09/2024 – Present] Postdoctoral fellow Université de Pau et des Pays de l'Adour
Identification of Bioactive molecules in Thermal Spring Waters for skin microbiome enhancement
CARUSO Project: Interreg POCTEFA
Institute of Analytical Sciences and Physico-Chemistry for the Environment and Materials (IPREM)
Institut des sciences analytiques et de physico-chimie pour l'environnement et les matériaux
Pau University (UPPA), **CNRS**, **France**
Advisors: Dr. Cristiana Cravo-Laureau, Prof. Robert Duran

[2023] Qualification- Section 68: Assistant Professor by the Ministry of Higher Education and Research, **France**

[02/2022 – 2024] Postdoctoral fellow
Modelling of chemical and microbiological processes in the atmosphere and soil
MOBIDIC Project: Make Our Planet Great Again (MOPGA)
Clermont-Ferrand Institute of Chemistry (ICCF)
Institut de Chimie de Clermont-Ferrand
Clermont Auvergne University (UCA), **CNRS**, **France**
Advisor: Dr. Barbara Ervens

[09/2018 – 12/2021] Doctoral student
Thesis: Anthropogenic impacts on microbial communities in agriculture arid zones
AQUASALT Project: ERANETMED Program
Institute of Experimental Biocenotics and Agrosystems (IBEAS)
Institut de biocénotique expérimentale et des agrosystèmes
Pau University (UPPA), CNRS, France
Advisors: Prof. Robert Duran, Dr. Christine Cagnon

III. Education

[09/2018 – 01/2022] Ph.D. in Physiology and Biology of organisms-Populations-Interactions
Pau University (UPPA), **France**

[09/2016 – 06/2018] MSc. in Management and Conservation of Natural Resources
Lebanese University, Beirut, **Lebanon**

[09/2013 – 06/2016] BSc. in Biology
Lebanese University, Beirut, **Lebanon**

IV. International Scientific stays

[06/2026 – 08/2026] Research training **(Confirmed)**
Metagenomic insights into microbial dynamics under redox conditions in extreme thermal waters
Host: Pr. Carmen Castañeda, **CSIC, Zaragoza, Spain**

[09/2024] International School on Ion Mobility MASS SPECTROMETRY
Ion Mobility mass spectrometry: Fundamentals, Advances, and Applications
Host: Ettore Majorna Foundation, **Erice, Italy**

[05/2022] **UNESCO-MAB Laureate**
Research In the World Network of Biosphere Reserve
Determine a relevant set of microbial bioindicators based on chemical and molecular approaches that can be used to assess the ecological status of Biosphere Reserve soils
Eberswalde University for Sustainable Development
Host: Biosphere Reserves Institute, **Eberswalde, Germany**

[10/2019 – 02/2020] **Doctoral Mobility**
Adapt microgAMBI microbial Index for contaminated soils and sediments
AZTI Marine research center, **Pasaia, Spain**
Hosts: Prof. Angel Borga, Dr. Anders Lanzén

V. Peer reviewed publications

Bourhane Z., Lanzén A., Cagnon C., Ben Said O., Mahmoudi E., Coulon F., Atai E., Borja A., Cravo-Laureau C., Duran R. (2022). **Microbial diversity alteration reveals biomarkers of contamination in soil-river-lake continuum**, *Journal of Hazardous Materials*.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. et al. (2023). **Vertical organisation of microbial communities in hypersaline wetland, Salineta,** *Frontiers in Microbiology*.

Amato P., Mathonat F., Nunez L., Péguilhan R., **Bourhane Z**., Rossi F., Vyskocil J., Joly M., Ervens B. (2023). **The aeromicrobiome: the selective and dynamic outer-layer of the Earth's microbiome,** *Frontiers in Microbiology*.

Bourhane Z., Amato P., Ervens B. (2025). **Comparison of polycyclic aromatic hydrocarbons (PAHs) biodegradation in the atmosphere and soil.** (Under review: Submitted to Atmospheric Chemistry and Physics)

Liu Bang., **Bourhane Z.,** Yoa Jun P., Castañeda C., Cagnon C., Cravo-Laureau C., Chen Z., Duran R. (2026). **Genome-resolved analyses disentangle the strategies of hypersaline microbiomes and metabolic interactions within a hydrological gradient**. (Under review: Submitted to Environmental Science &Technology)

Aziza Kouchou, Fadwa Tadist, Zeina Bourhane, Olfa Ben Said, Naoual Rais, Robert Duran, Naïma El Ghachtouli· **Effects of pollution on the activity of the soil/sediment microbial communities from a soil-river-lake continuum in Ichkeul Lagoon**

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. **Metagenomic insights into microbial dynamics under redox conditions in a hypersaline wetland soil using Indicator of Redox in Soils (IRIS) films**

Zeina Bourhane, Robert Duran, Cristiana Cravo-Laureau. **Unveiling the metagenomic profiles of Pyrenean thermal waters**

VI. Oral presentations

Bourhane Z., Cravo-Laureau C., Meselier M., Pinaly H, Duran R., Preud'homme H. *Are bioactive molecules in thermal waters revolutionizing cosmetic products?*. **AFEM:** 12th Francophone Association of Microbial Ecology Congress, Beaune, France, 18-21 November **2025**. (Accepted)

Bourhane Z., Amato P., Ervens B. *A comparison of the loss of polycyclic hydrocarbons (PAHs) in soil and in the atmosphere.* **European Geosciences Union (EGU**), Vienna, Austria, 23–28 April **2023**.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Microbial diversity alteration reveals biomarkers of contamination in soil-river-lake continuum.* **International Conference: Science and Research in, for and with UNESCO Biosphere Reserves**, Eberswalde, Germany, 16-20 May **2022**. ***(Invited Speaker)***

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Role of microbial diversity alterations in revealing biomarkers of contamination in soil-river-lake continuum.* **EcotoxicoMicYR**, December **2021**. ***(Invited Speaker)***

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Microbial diversity alteration reveals biomarkers of contamination in soil-river-lake continuum.* **PhD students Journey**, Pau University, Pau, France, 8-9 July **2021**.

Bourhane Z., Lanzén A., Cagnon C., Ben Said O., Mahmoudi E., Coulon F., Borja A., Cravo-Laureau C., Duran R. *Bacterial bioindicators for reporting contamination in a coastal soil/river/lake continuum*. **SETAC Europe 31st Annual Meeting**, Seville, Spain, 3-6 May **2021**.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Role of bioindicators in the environmental assessment.* AZTI, Marine Research, Basque Research and Technology Alliance (BRTA). Pasaia - Gipuzkoa, Spain, Nov. **2019**.

Bourhane Z., Cravo-Laureau C., Duran R. Microbial biodiversity in saline shallow lakes. Estacion Experimental de Aula Dei, CSIC. Zaragoza, Spain, March **2019**. ***(Invited Speaker)***

VII. Poster presentations

Bourhane Z., Cravo-Laureau C., Meselier M., Pinaly H, Duran R., Preud'homme H. *Novel Bioactive molecules in Thermal Waters: A New Frontier in Cosmetic Innovation.* **EcotoxicoMicYR**, October-November **2025.**

Bourhane Z., Cravo-Laureau C., Meselier M., Pinaly H, Duran R., Preud'homme H. *Bioactive molecules in thermal waters: A revolution in cosmetic products.* **Microbes:** 20th Congres French Society of Microbiology Bordeaux, France, 24-26 September **2025**.

Bourhane Z., Chevillon V. *Bioactive molecules identification in POCTEFA area plant extracts and thermal spring waters for skin enhancement microbiome in biosourced cosmetic.* **International School on Mass Spectrometry**, Erice, Italy, 9-14 September **2024**.

Bourhane Z., Cagnon C., Castañeda C., Rodríguez-Ochoa R., Álvaro-Fuentes J., Cravo-Laureau C., Duran R. *How microbial communities organize vertically in hypersaline wetland to ensure biogeochemical cycles?.* **EcotoxicoMicYR**, December **2023**.

Bourhane Z., Ervens B. *Comparison of bacterial biodegradation of polycyclic aromatic hydrocarbons (PAHs) in the atmosphere and soil*. **International Society for Microbial Ecology** (**ISME18**), Lausanne, Swiss, 14-19 August **2022**.

Bourhane Z., Cagnon C., Castañeda C., Rodríguez-Ochoa R., Álvaro-Fuentes J., Cravo-Laureau C., Duran R. *Vertical organisation of microbial communities in Salineta hypersaline wetland, Spain*. **FEMS Conference on Microbiology**, Belgrade, Serbia, 30-2 july **2022**.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Bacterial bioindicators for reporting contamination around Ichkeul Lake*. **EcotoxicoMic 2020**, Second International Conference on Microbial Ecotoxicology, Montpellier, France, 6-9 October **2020**.

Bourhane Z., Cagnon C., Cravo-Laureau C., Duran R. *Anthropogenic impact on microbial communities in agriculture arid zones*. **PhD students Journey**, Pau University, Pau, France, 3-6 June **2019**.

Research grants and Awards

2026: Research grant. FEMS (Federation of European Microbiological Societies) **(3000 €)**
2023: Travel grant. AFEM Young researcher mobility **(1200 €)**
2022: Travel grant. FEMS Conference attendance Grant, Belgrade **(1200 €)**
2021: Award. Laureate of the **UNESCO MAB for young scientists (5000 $)**
2021: Conference Attendance grant. AFEM Young researcher mobility **(500 €)**
2019: Travel research grant. Grant of Excellence, Energy and Environment Solutions (E_2S), France **(5000 €)**
2019: Travel research grant. Aquitaine Mobility (AquiMob) International Mobility Grant, UPPA **(1200 €)**
2018-2021: PhD scholarship AQUASALT ERANET MED (Number NMED-0003-01) **(98 500 €)**

Courses and Certificates

Developing Skills for Women Leadership in Climate Action
Developed by UNITAR and C40 with the exclusive support of the Fondation L'Oréal, and with knowledge contributions from Columbia Climate School and CARE France.

Nature-based Solutions for Disaster and Climate Resilience
Organised by: UN Environmental Program, PEDRR, EU, and SDG Academy.

Membership of scientific organizations

2026- Pins of Sciences (Event Mananger**)**
2026- ISME member (International Society for Microbial Ecology)
2025- SFM member (Société Française de Microbiologie) : https://www.sfm-microbiologie.org/
2025- NextGen4MED: Network of young researchers focused on sustainable water management in the Mediterranean
2024- SFBI member (Société savante de la bioinformatique) : https://www.sfbi.fr
2020- EcotoxicoMic member: https://ecotoxicomic.org
2019- AFEM member (Francophone Association of Microbial Ecology)

Peer reviewer

Frontiers in microbiology, Environnemental Pollution, Microbial Ecology, Environmental Research, Scientific Reports, Thalassas (Springer)

Technical skills

Bacteria culture
DNA extraction and PCR
Agarose gel electrophoresis
Statistics and bioinformatics: QIIME2, R Studio
Modelling: quantitative prediction
Mass Spectrometry

X. Communication and Vulgarization

A- School

1- Village des Sciences

Collège *Tristan Derème – Oloron (la ou j'ai suis allee)*

Lycée Jules Supervielle /OLORON-SAINTE-MARIE (note sur site uppa)

Filles et sciences : atelier sur les stéréotypes et comment s'en prémunir, en prenant l'exemple principal des filles dans les sciences et techniques

https://www.univ-pau.fr/fr/agenda/saps/fete-de-la-science-village-des-sciences-oloron-sainte-marie.html

2- Circuit Scientifique

Lycée 2: Cité Scolaire Gaston Crampe -Aire sur Adour

Table ronde : Devenir professeur-chercheuse en microbiologie

https://partage.univ-pau.fr/service/home/~/?auth=co&loc=fr&id=95332&part=2

Atelier : Micro-organismes amis ou ennemis ?

https://partage.univ-pau.fr/service/home/~/?auth=co&loc=fr&id=95332&part=3

https://www.univ-pau.fr/fr/agenda/saps/fete-de-la-science-circuit-scientifique-hors-les-murs-oloron-et-aire-sur-l-adour.html

3- Les Cordées Scientifiques

Collège Joseph Peyré - Garlin

Atelier : Le rôle des micro-organismes dans l'environnement (15/01/2026)

Collège Jean Sarrailh - Monein

Atelier : Le rôle des micro-organismes dans l'environnement (20/01/2026)

Collège d'Arzacq

Atelier : Les Marvels de l'environnement (03/03/2026)

B- Workshops

1- International Microorganism Day (IMD25)

https://www.internationalmicroorganismday.org/blog/imd2025-france

2- Pint of Sciences 2026

https://pintofscience.fr/equipe/Pau/

C- Blogs

1- Panorama

https://panorama.solutions/en/solution/harnessing-hidden-power-thermal-waters-microbiomes-healthy-planet

https://sdgacademy.org/thermal-waters-from-research-to-innovation/?utm_content=buffer93014&utm_medium=social&utm_source=facebook.com&utm_campaign=buffer

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D- Ecotoxicomic YR 2025

https://www.youtube.com/watch?v=jd-LUYRIm28

Grants E2S

file:///E:/zbourhane/Downloads/Laureates_InternationalDoctoralMobility_session2_2019.pdf

https://iprem.univ-pau.fr/fr/collaborations/projets-de-recherche/caruso-2024-27/etude-des-micro-organismes-dans-les-eaux-thermales-des-pyrenees-un-potentiel-cache-pour-des-cosmetiques-durables.html?mtm_campaign=Newsletters/Le%20Mensuel%20de%20l%27Universit%C3%A9&mtm_medium=email&mtm_source=N%C2%B085%20-%2007/10/2025%20-%20ERRATUM%20-%20Lisez%20le%20Mensuel%20de%20l%27universit%C3%A9-85-2025-10-07